Felix Zhang

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Education

University of Toronto Sept. 2025 – Jan. 2027
Master of Science in Computer Science *4.0/4.0 cGPA*
Advisor: Prof. Qizhen Zhang

University of Toronto Sept. 2021 – June 2025
Honours Bachelor of Science in Computer Science; Major in Mathematics *3.96/4.0 cGPA*

Publications

PD3: Prefetching Data with DPUs for Disaggregated Memory May 2026
Sidharth Sankhe, **Felix Zhang**, Umayrah Chonee, Sherman Lim, Jason Hu, Jialin Li, Qizhen Zhang
23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI '26)

Research Experience

Research Assistant; Far Data Lab, University of Toronto Sept. 2024 – Aug. 2024
- Parallelized the execution of *Monodepth2* in **Python** and **C++** with the latter achieving linear performance scaling with the number of threads
- Built a DPU-based prefetcher *PD3* with a team of 6 which intercepts network traffic to predict and prefetch data for tiered key-value stores
- Designed an external service for offloading shuffle operations from database management systems which supports either a memory or storage backend managed by DPUs

Research Assistant; University of Toronto March 2024 – Dec. 2024
- Worked on a team of 11 to implement a pedagogical kernel *KidneyOS* in **Rust** to be used in an introductory operating systems course with **500+** students annually
- Enabled thread creation and destruction, multi-threading, pre-emptive scheduling within the thread system
- Led a team of 3 to implement POSIX-compatible syscalls and add support running user-space executables

Industry Experience

ML Runtime Engineer; Cerebras Systems May 2024 – Aug. 2025
- Implemented a runtime virtual memory system in **C++** with a team of 3 which pre-emptively loads data before it is accessed, allowing off-chip memory to be used for the first time with only a **10%** performance penalty
- Added support for network storage in the paging system with remote direct memory access, providing **100** GB/s read and write speeds with **10** μs latency to multiple remote servers
- Enabled the ability log and replay the network operations, decreasing the time to recreate stalls and timeouts by over **80%**, and setup unit tests to automatically catch breakages and performance regressions in the network layer
- Improved the throughput of the network layer by **6%** when transferring data by implementing best practices for remote direct memory access and reducing setup overhead
- Determined the cable and port mapping for a one node cluster which will be used for all future deployments

Awards & Scholarships

Ontario Graduate Scholarship ($15 000), University of Toronto May 2025 – April 2026
Queen Elizabeth II Graduate Scholarships in Science and Technology ($15 000) (Declined), University of Toronto May 2025 – April 2026
Fields Undergraduate Summer Research Program ($3 800), Fields Institute June – Aug. 2024
Louis Savlov Scholarship in Sciences And Humanities ($1 000), University of Toronto Nov. 2023 – Jan. 2025
Dean's List Scholar, University of Toronto Jan. 2022 – June 2025
Second Malcom Wallace Scholarship ($5 000), University of Toronto Sept. 2021 – Oct. 2024
University of Toronto Scholar ($7 500), University of Toronto Sept. 2021

Technical Skills

Languages C++, Rust, Python, C, C#, Bash, Java
Frameworks Catch2, PyTest, PyTorch, TensorFlow, OpenCV, scikit-learn, NumPy, Pandas, SciPy, Eigen3
Tools Git, Linux, Unix, tmux, CMake, Docker, Anaconda, Google Colab, QEMU, Jupyter, WSL, Slurm